Final Report

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Part IIA Project: *Neural Control with Adaptive State Estimation* (GG4)
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Introduction

We are given a Brain and a Brain–Machine Interface (BMI), which controls an arm. We are tasked with controlling the position of the hand at the end of the arm. The Brain takes $n_u = 2$ inputs and produces $n_y = 16$ outputs; we assume it is a Linear Gaussian State–Space Model (LGSSM). The BMI takes the 16-dimensional observations from the Brain and applies a fixed linear decoder to map them to 2 latent states: x_1, the muscle selection rhythm; and x_2, power rhythm. The BMI stores a rolling history of x_1, which is passed through 4 fixed band–pass filters which measure how strongly x_1 oscillates at that muscle's defined excitation frequency. A history of x_2 is also stored in state, and the current value is compared to a half– and full–period in the past: if strong oscillation is found, evidence accumulates; otherwise, evidence decays, producing a smooth power state. Muscle activation is the product of activation and the selection vector. Each muscle produces force proportional to its activation, and for each joint, the net force is the sum of signed muscle forces. Positive forces represent flexion; negative forces represent extension. A dry-friction threshold must be overcome, and once this is overcome, the remaining net force (once subtracting the dry friction) is divided by wet friction representing viscous resistance. Joint angles are integrated, and 2-link arm kinematics give the hand position.

Rules of engagement

The holistic view of this project is to be able to reason and interpret a controller which takes observations from a brain (e.g. via electrodes on the scalp) to move a prosthetic arm. Critically, the brain is a black box and its model "parameters" cannot be directly observed. In the spirit of this view, we propose specific rules of engagement surrounding our analysis. We do not attempt to decompile the provided Brain executables to derive the system matrices. We do not average over multiple Brain seeds, removing noise to derive system matrices, we reserve this for evaluation purposes only to demonstrate robustness against noise. Where we do run experiments that are not physically realisable, we do this to demonstrate the direction of our analysis, and we replicate results in empirically valid ways alongside. We do not directly use the weights of the BMI in our control efforts: we allow ourselves to read their values for experimental design, but any results we must prove empirically before use. We only allow the use of the 16-dimensional observations from the Brain and the hand position given by the BMI for closed-loop control; we do not rely on the internal state of the BMI (e.g., arm angles or latent state variables) for feedback. We also do not expect or attempt to control the elbow joint past 180° to mimic the capabilities of an anatomically accurate arm. We motivate these constraints as to achieve resilience and independence to both: inter-individual variation in brains (and therefore system matrices); and manufacturers of prosthetic arms (and therefore BMI decoding, state, and arm kinematics defined by model weights).

System identification

As we assume the Brain is a LGSSM, we explore system identification as a means to understand the system matrices defining the model:

$$\begin{aligned} \underline{x}_{t+1} &= \underline{A}\,\underline{x}_t + \underline{B}\,\underline{u}_t + \underline{w}_t, & \underline{w}_t &\sim \mathcal{N}(\underline{0}, \underline{Q}) \\ \underline{y}_t &= \underline{C}\,\underline{x}_t + \underline{o}_t, & \underline{o}_t &\sim \mathcal{N}(\underline{0}, \underline{R}) \end{aligned}$$

Identification evaluation

To evaluate our estimates, we have two strategies: the reconstruction of observations and the reconstruction of frequency response. To measure the reconstruction of observations, we use the coefficient of determination defined with the Frobenius norm:

$$R^2 = 1 - \frac{\left\| Y - \widehat{Y} \right\|_F^2}{\left\| Y - \overline{Y} \right\|_F^2}$$

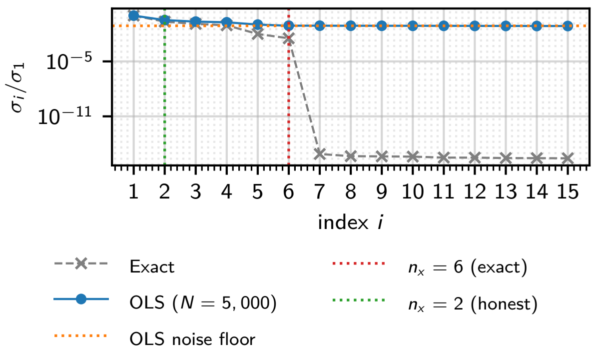
This form is chosen for its effectiveness at aggregating errors across all time points and all observation dimensions. We weight this up against its disadvantages, which include ignorance of observation covariance and temporal correlations. To measure the reconstruction of frequency response, we inspect a frequency map derived analytically from estimates of system matrices, and compare characteristics with empirical results from a a single run of the Brain. We find the transfer function of the LGSSM, dependent on only the deterministic components (matrices $\underline{A}$, $\underline{B}$, and $\underline{C}$):

$$\underline{H}(z) = \underline{C}\left(z\underline{I} - \underline{A}\right)^{-1}\underline{B}$$

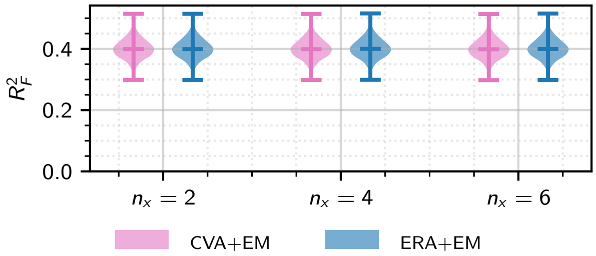
We define an estimate to reconstruct the frequency response well if it can minimise the error between analytic and empirical values of $|H(e^{j\omega})|$ across all neurones and frequencies $[0, \pi]$. Given the emphasis on rhythm in the BMI we weigh frequency response reconstruction more than observation reconstruction if a compromise is necessary.

Identification methods

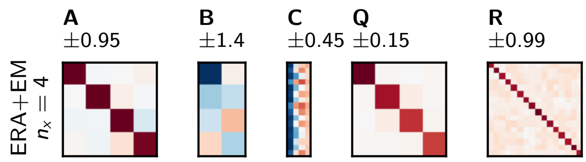
We run two `Brain` instances with and without input to isolate and subtract the noise. We can then calculate the exact Markov parameters, construct Hankel matrices, and form a singular-value spectrum. With this analysis we find σ_i/σ_1 fall after $i = 6$, indicating the true latent dimension number to be $n_x = 6$ (Figure **??**). However, this method is not physically realisable: you cannot run the same physiological brain twice with the same random state. This would mean rewinding a brain, re-run a trial without a stimulus, and subtract: noise is truly stochastic outside of a simulation. A physically realisable single-run Ordinary Least Squares (OLS) regression estimate uncovers a noise floor that obscures modes 3–6. Balancing likelihood and model complexity with Bayesian Information Criterion (BIC) model selection, we identify $n_x = 2 \neq 6$ (Figure **??**).



We plot a heatmap of $|H(e^{j\omega})|$ and find that Canonical Variate Analysis (CVA) and Expectation-Maximisation (EM) to reconstruct u_1 well, but not u_2 over all n_x candidates $\{2, 4, 6\}$ (Figure **??**, N.B. only $n_x = 4$ plotted). We turn to an estimation method that uses the Eigensystem Realisation Algorithm (ERA) and EM that utilises OLS-estimated Markov parameters and find similar results for $n_x = 2$: good reconstruction of u_1 but poor for u_2. However, in contrast to CVA+EM, reconstruction of the u_2 frequency response improved with $n_x = 4$ and $n_x = 6$ (Figure **??**). More generally, we observe that the `Brain` acts as a low-pass filter. The observation reconstruction quality remained similar between CVA+EM and ERA+EM (Figure **??**).



As there was not a huge difference between $n_x = 4$ and $n_x = 6$, we select ERA+EM and $n_x = 4$, which yields system matrices visualised in Figure **??**.



Closed-loop spectral control of brain observations

We test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population mean activity. We motivate the following evaluation framework and method with the emphasis on controlling rhythms in the `Brain` observations for influence over BMI latent variables and the arm downstream.

Spectral control evaluation

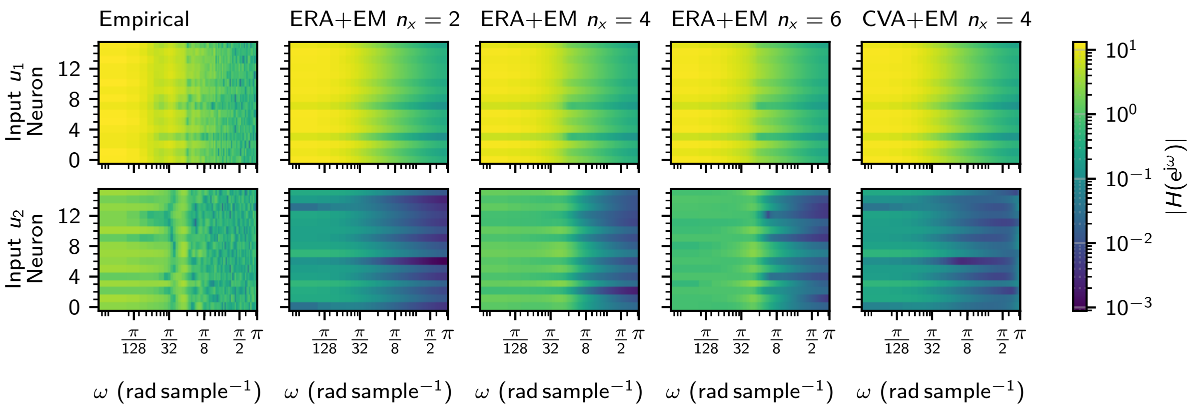
We use the following metrics to evaluate each controller: spectral Root Mean Square (RMS) error, which measures amplitude deviation at target frequencies; the maximum amplitude error, the worst-case per-component deviation; and the off-target power ratio, the mean spectral amplitude at non-target frequencies, relative to mean target amplitude.

Spectral control method

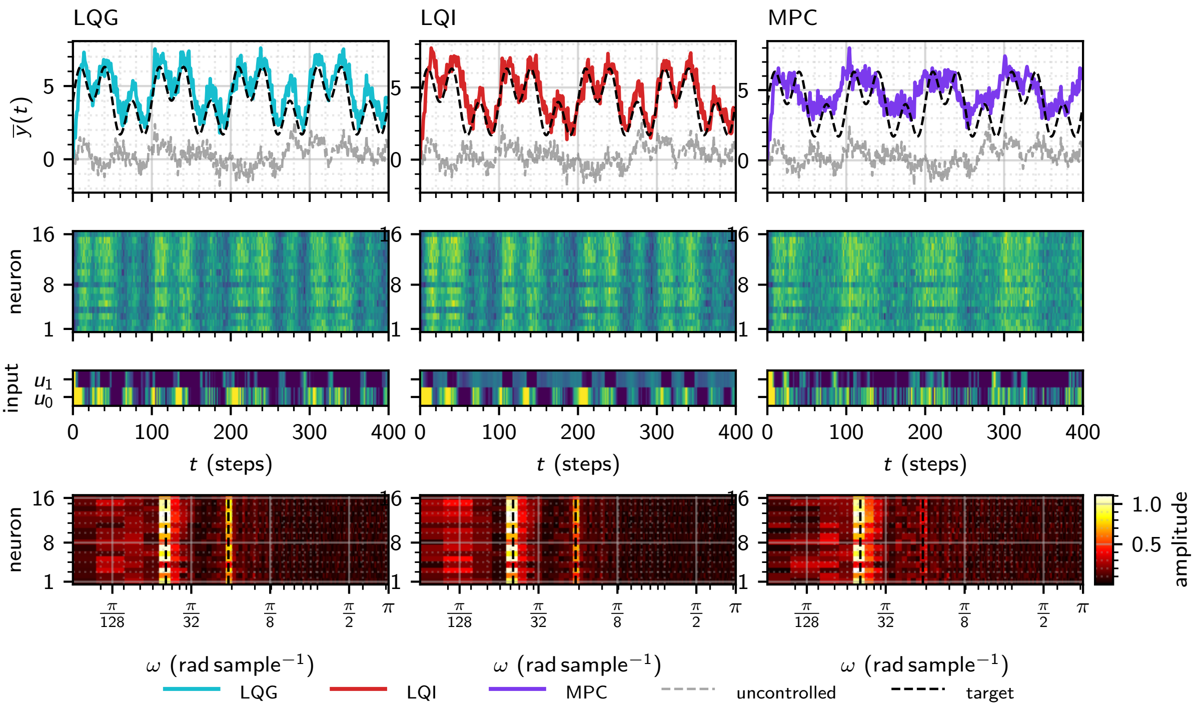
We offer 3 candidates for control strategies: Linear Quadratic Gaussian (LQG) to respect the `Brain`'s LGSSM structure; Linear Quadratic Integral (LQI) to correct steady-state error; and Model Predictive Control (MPC) to optimise for and appreciate constraints on $u_i \in [0, 1]$.

Spectral control results

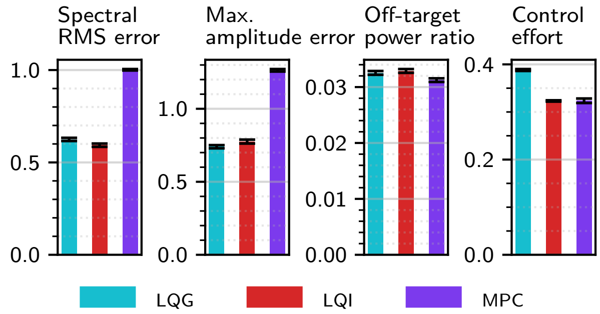
Inspecting the raw population mean traces, we find LQG overestimates the population mean with a steady-state error, which is removed effectively by LQI. MPC fails to reconstruct the higher frequency components. We select LQI due to its lower control effort compared to LQG and better reconstruction performance over MPC.

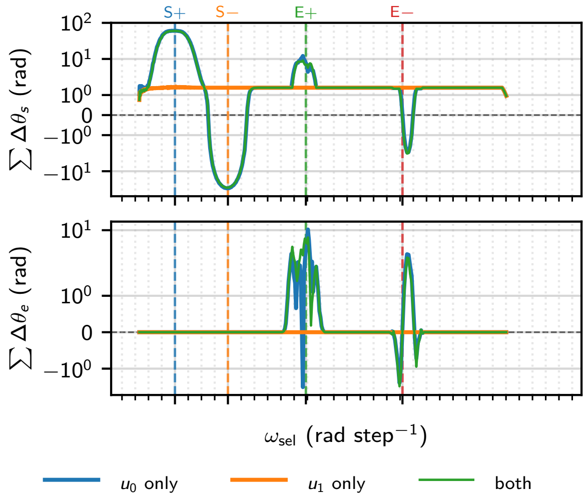


Empirical
ERA+EM $n_x = 2$
ERA+EM $n_x = 4$
ERA+EM $n_x = 6$
CVA+EM $n_x = 4$
Input u_1 Neuron
Input u_2 Neuron
ω (rad sample^{-1})
$|H(e^{j\omega})|$



LQG
LQI
MPC
$\overline{y}(t)$
neuron
input
t (steps)
ω (rad sample^{-1})
amplitude
LQG
LQI
MPC
uncontrolled
target





1 Hand control

The mechanism of the BMI of muscle selection and a shared power state means that simultaneous control of different muscles is unlikely, as there is no granular control to direct different muscles at different power at the same time. This constrains our control strategy to be a sequential one, directing one muscle at a time.

Evaluating hand control

We aim to minimise the final distance to target, the Euclidean distance between the hand and the target at the end of a trial. We define a success threshold at 10. We aim to maximise the reach rate, the fraction of trials that reach within this success threshold. This metric captures reliability across diverse targets in the reachable area: a controller with low mean final distance distance but occasional failures is not practically useful for a prosthetic. We minimise the time to threshold, the number of steps until the hand first enters the distance success threshold: a useful prosthetic should converge in reasonable time to provide utility. We maximise the sector fraction, the proportion of time the hand spends in the optimal sector: an angular wedge containing the target and the start position. We minimise the rotational distance, the total angle swept by the hand trajectory over the trial. These two metrics penalise inefficient trajectories (e.g. looping behaviours), which would be inconvenient and unpredictable for a prosthetic. We minimise the control effort, the sum of square brain inputs, in an effort to make inputs more interpretable and efficient.

Muscle selection

We sweep frequencies for both inputs and measure shoulder and elbow angles, determined by inverse kinematics to find the specific frequencies that excite specific muscles. We empirically recover the frequency bands defined in the BMI weights using a single seed, shown in Table **??** and Figure **??**.

We observe that amplitude decreases with frequency, making it harder to drive muscles that require higher frequencies. This may be due to the low-pass action of the `Brain`. We also observe cross-talk for elbow movements but not shoulder movements: the shoulder can be moved independently of the elbow, but the elbow cannot be moved indepedently of the shoulder.

We then ask whether neural observations carry a muscle-specific spatial code: if different neurons respond preferentially at each selection frequency, the control target could be weighted to favour one muscle over another. For each selection frequency we record the full $n_y = 16$ D amplitude spectrum and normalise the per-neuron profile by its own neuron-mean (Figure **??**).

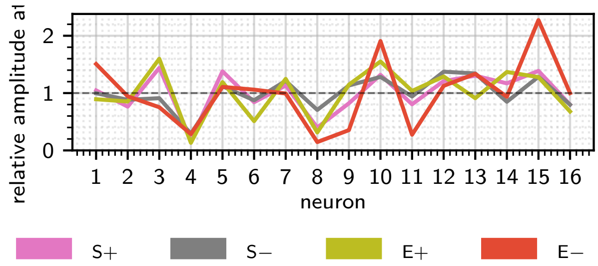
Muscle identity is therefore encoded temporally by the oscillation frequency of x_1, not spatially by which neurons are active (Figure **??**). No differential weighting of neurones in the spectral control target offers muscle selectivity beyond what the BMI's bandpass filters already provide. This justifies controlling the population mean at the spectral target.

Control method

The core pipeline is a 3 step process. First is band-identification, an open-loop sinusoidal sweep over both brain input channels, with cumulative shoulder and elbow displacement (inferred via inverse kinematics) recorded per frequency. The four bands are identified, which correspond to S+, S−, E+, and E−. The next step is gain calibration, where a short-closed loop LQI trial at each band frequency targets a sinusoidal spectral setpoint, and mean joint angle per unit amplitude forms a 2×4 gain matrix where rows are joints, and columns are bands. The third is the main control loop, where inverse kinematics converts cartesian target to a joint-angle goals, band amplitudes are solved from the gain matrix and an LQI controller optionally

Muscle	ω^*_{true} (rad step^{-1})	ω^*_{rec} (rad step^{-1})	$\Delta\omega$ (rad step^{-1})
S+	0.4378	0.4263	−0.0115
S−	0.7854	0.7800	−0.0054
E+	1.2875	1.3006	0.0131
E−	1.9764	1.9458	−0.0306

Source



closes the loop on the population mean.

The elbow to shoulder cross-talk motivates a control strategy that first achieves the correct elbow angle, then reduces control gain over the elbow, and focuses on using S+ and S− to correct the shoulder angle. In addition, during the initial correction angle phase, we should alternate between correcting the elbow angle and correcting the shoulder angle, as attempting to move the elbow moves the shoulder. To mitigate cross-talk further, we drive the shoulder in the opposite direction of the expected cross-talk.

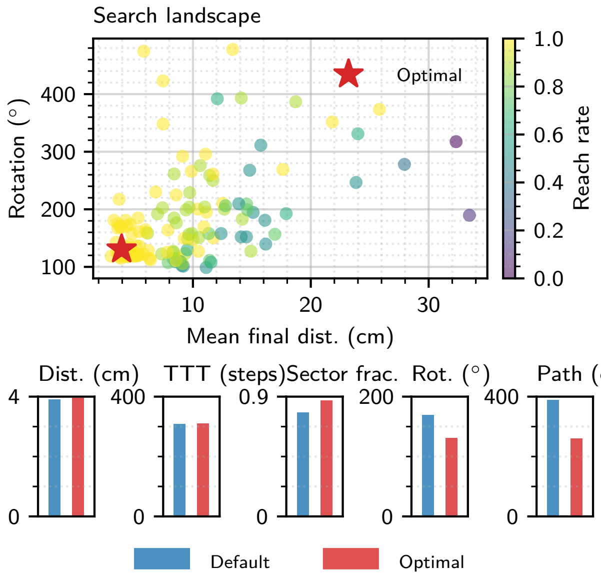
We always begin with targeting elbow error first. Each elbow/shoulder phase has a minimum length and maximum length. In addition, when controlling the elbow, there is a maximum shoulder drift, such that if upon moving the elbow the shoulder drifts too much, we switch to controlling the shoulder. We also provide per-band gain multipliers applied to the gain matrix values before the solve, to further tune individual band aggressiveness.

Hyperparameter optimisation

With so many model parameters defined by the complex sequential controller design, we turn to hyperparameter optimisation to search the parameter space for an optimal solution. We use Tree-structured Parzen Estimator (TPE) via the `optuna` library to run two phases: first, a large set of Bayesian Optimisation (BO) trials; then the top trials are re-evaluted on a full target grid, and the best is selected. We define an objective score:

$$\mathcal{S} = \overline{d}_{\text{final}} - 20\,\overline{f}_{\text{sector}} - 10\,\overline{p}_{\text{reached}} + 50\,\overline{\max(0,\ \theta_{\text{rot}} - 2\pi)}$$

This directly penalises final distance, inefficient trajectories, and rewards successful reach rate to create a more interpretable optimisation process. Figure **??** shows how the optimisation process did not reduce final distance or the time to threshold, but instead made the trajectory more sensible, increasing the sector fraction and reducing the rotation.



Results

The most optimal controller's metrics are summarised in Table **??**, its losses over time are plotted in **??**, an example trajectory is illustrated in Figure **??**, and a heatmap of targets and the final distances to them is drawn in Figure **??**.

Metric	Default (mean ± SD)	Optimal (mean ± SD)
Final dist. (cm)	3.90 ± 2.76	3.95 ± 1.42
Reach rate	1.000 ± 0.000	1.000 ± 0.000
TTT (steps)	308 ± 155	309 ± 147
Sector frac.	0.782 ± 0.126	0.871 ± 0.082
Rot. dist. (°)	169.3 ± 121.8	130.6 ± 65.5
Path length (cm)	389.4 ± 211.3	259.6 ± 68.4

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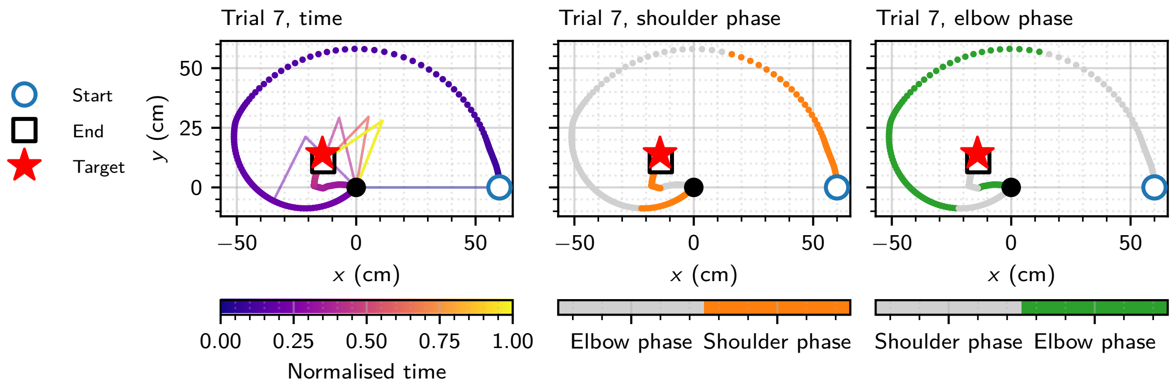

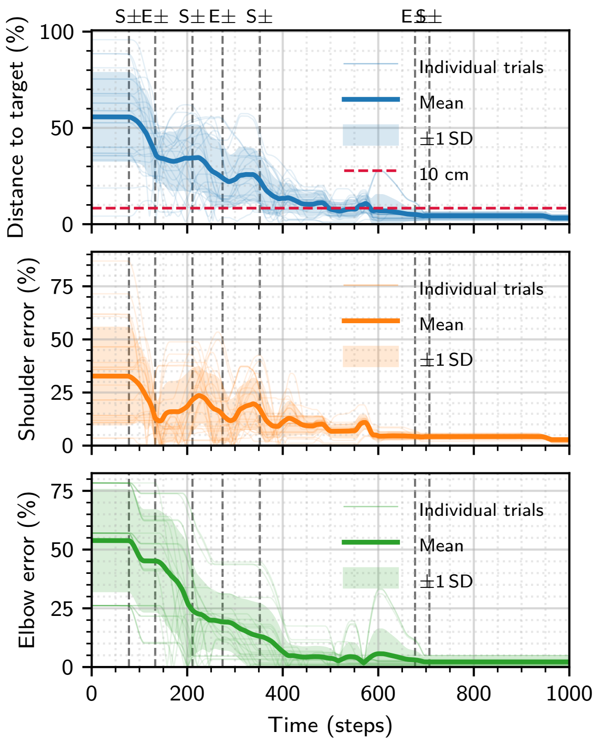

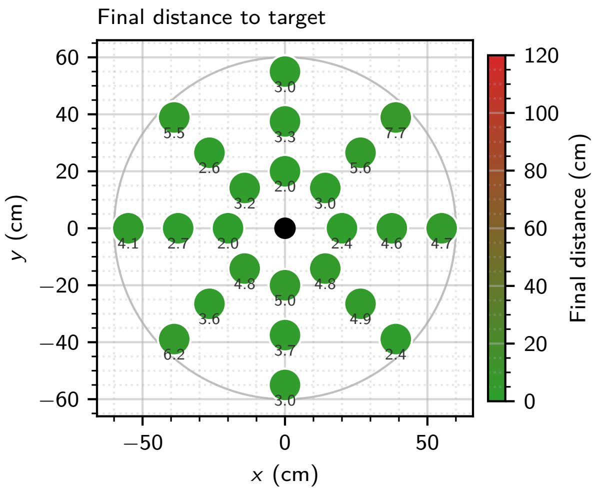

Sensitivity to model parameters

We carry out ablation analysis, varying one parameter of the controller in isolation and holding everything else fixed to measure how sensitive performance is to that parameter. Hyperparameters under analysis include: the elbow band scale, how much the elbow gain is turned down relative to the shoulder gain Figure **??**; the blend factor, how much we compensate for cross-talk by driving the shoulder in the opposite direction Figure **??**; closed-loop vs open-loop control Figure **??**; and whether we constrain the elbow to a maximum of 180° or not Figure ??. We filter the metrics using a one-way Analysis of Variance (ANOVA), run independently for each metric across all conditions, with a null hypothesis that all group means are equal. A metric is plotted only if the ANOVA p-value satisfies $p \leq 0.05$.